

% Analog signal to Digital signal

```
clear all
```

```
clc
```

```
Fs = 150;
```

```
T=1/Fs;
```

```
bit=2;
```

```
dt = 1/Fs;
```

```
StopTime = 0.025;
```

```
t=0:0.00001:StopTime;
```

```
n=0:1:(StopTime/dt);
```

```
Fc =100;
```

```
Ya =5*sin(2*pi*Fc*t)+3*cos(2*pi*6*Fc*t);
```

```
y_max=max(Ya);
```

```
y_min=min(Ya);
```

```
y =5*sin(2*pi*(Fc/Fs)*n)+3*cos(2*pi*6*(Fc/Fs)*n);
```

```
ns=length(y);
```

```
q_out=zeros(1,ns);
```

```
del=(y_max-y_min)/(2^bit);
```

```
low=y_min+(del/2);
```

```
high=y_max-(del/2);
```

```
for h=low:del:high
```

```
    for b=1:ns
```

```
        if(((h-del/2)<y(b)) && ((y(b)<=(h+del/2))))
```

```
            q_out(b)=h;
```

```
        end
```

```
    end
```

```
end
```

```
q_error=q_out-y;
```

```
r_memory=bit*ns;  
fprintf('Required memory: %d bit',r_memory);
```

```
subplot(321)  
plot(t,Ya,'r','Linewidth',1);  
xlabel('Time');  
ylabel('amplitude');  
title('Analog Signal');
```

```
subplot(322)  
stem(n,y,'c','Linewidth',1);  
xlabel('sample number');  
ylabel('amplitude');  
title('Discrete time continuous valued Signal');
```

```
subplot(323)  
stem(n,q_out,'m','Linewidth',1);  
xlabel('Sample number');  
ylabel('amplitude');  
title('Digital (discrete time discrete valued) Signal');
```

```
subplot(324)  
plot(n,q_error,'b','Linewidth',1);  
xlabel('Sample Number');  
ylabel('amplitude');  
title('Quantization Noise');
```

```
subplot(325)  
plot(n.*T,q_out,'g','Linewidth',1);  
xlabel('Time');  
ylabel('amplitude');
```

```

title('Reconstructed Analog Signal');

subplot(326)
plot(n.*T,q_out,'g','Linewidth',1);

hold on

plot(t,Ya,'r','Linewidth',1);

title('Discrete time continuous valued Signal');

xlabel('Time');

ylabel('amplitude');

title('Reconstructed Analog Signal Vs Analog Signal');

```

.....

%Shifting

```

clc;

clear;

close all;

%Original signal
x=[1 3 5 7 9];

n=-2:2;

%Shifting amount

k=2;

%Left shift:  $y[n]=x[n+k]$ 

n_left=n-k;

%Right shift:  $y[n]=x[n-k]$ 

n_right=n+k;

%Plot signal

figure;

%Original signal

subplot(3,1,1);

stem(n,x,'filled','LineWidth',1.5);

grid on;

title('Original Signal x[n]');

```

```

xlabel('n');

ylabel('Amplitude');

%Left shift

subplot(3,1,2);

stem(n_left,x,'filled','LineWidth',1.5);

grid on;

title('Left Shift by 2:x=[n+2]');

xlabel('n');

ylabel('Amplitude');

%Right Shift

subplot(3,1,3);

stem(n_right,x,"filled",'LineWidth',1.5);

grid on;

xlabel('n');

ylabel('Amplitude');

```

%Folding

```

clc;

clear;

close all;

%Original Signal

x=[1 3 5 7 9];

n=-2:2;

%Folding(Time Reversal)

y=fliplr(x);

n_fold=-fliplr(n);

%Plot Original Signal

subplot(2,1,1);

stem(n,x,'filled','LineWidth',1.5);

grid on;

title('Original Signal x[n]');

```

```

xlabel('n');

ylabel('Amplitude');

%Plot Folded Signal

subplot(2,1,2);

stem(n_fold,y,'filled','LineWidth',1.5);

grid on;

title('Folded Signal  $y[n]=x[-n]$ ');

xlabel('n');

ylabel('Amplitude');

```

.....

%Even and Odd Part

```

clc;

clear;

close all;

%Original Signal

x=[2 4 6 8 10];

n = -2:2;

%Folded signal  $x[-n]$ 

x_fold = fliplr(x);

%Even and Odd part

x_even = (x+x_fold)/2;

x_odd = (x-x_fold)/2;

%Display values

disp('original Signal  $x[n]=$ ');

disp(x);

disp('Folded Signal  $x[-n]=$ ');

disp(x_fold);

disp('Even part  $x_e[n]=$ ');

disp(x_even);

disp('Odd part  $x_o[n]=$ ');

disp(x_odd);

```

%Plotting

figure;

%Original Signal

subplot(2,2,1);

stem(n,x,'filled','LineWidth',1.5);

grid on;

title('Original Signal $x[n]$ ');

xlabel('n');

ylabel('Amplitude');

%Folded Signal

subplot(2,2,2);

stem(n,x_fold,'filled','LineWidth',1.5);

grid on;

title('Folded Signal $x[-n]$ ');

xlabel('n');

ylabel('Amplitude');

%Even part

subplot(2,2,3);

stem(n,x_even,'filled','LineWidth',1.5);

grid on;

title('Even Part $x_e[n]$ ');

xlabel('n');

ylabel('Amplitude');

%Odd Part

subplot(2,2,4);

stem(n,x_odd,'filled','LineWidth',1.5);

grid on;

title('Odd Part $x_o[n]$ ');

xlabel('n');

```
ylabel('Amplitude');
```

.....

```
%Convolution
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Input signals
```

```
x = [1 2 3];
```

```
h = [1 2 1];
```

```
% Perform convolution
```

```
y = conv(x, h);
```

```
% Display results
```

```
disp('Signal x[n] =');
```

```
disp(x);
```

```
disp('Signal h[n] =');
```

```
disp(h);
```

```
disp('Convolution y[n] = x[n] * h[n]');
```

```
disp(y);
```

```
% Define indices
```

```
nx = 0:length(x)-1;
```

```
nh = 0:length(h)-1;
```

```
ny = 0:length(y)-1;
```

```
% Plot signals
```

```
figure;
```

```
% Input signal x[n]
```

```
subplot(3,1,1);
```

```
stem(nx, x, 'filled', 'LineWidth', 1.5);
```

```
grid on;
```

```
title('Input Signal x[n]');
```

```
xlabel('n');
```

```

ylabel('Amplitude');

% Input signal h[n]

subplot(3,1,2);

stem(nh, h, 'filled', 'LineWidth', 1.5);

grid on;

title('Input Signal h[n]');

xlabel('n');

ylabel('Amplitude');

% Convolution result

subplot(3,1,3);

stem(ny, y, 'filled', 'LineWidth', 1.5);

grid on;

title('Convolution Result y[n] = x[n] * h[n]');

xlabel('n');

ylabel('Amplitude');

```

%Correlation

```

clc;

clear;

close all;

% Input signals

x = [1 2 3];

y = [1 2 1];

% Cross-correlation

r = xcorr(x, y);

% Display results

disp('Signal x[n] =');

disp(x);

disp('Signal y[n] =');

disp(y);

disp('Cross-Correlation r_xy[l] =');

```



```

disp(r);

% Define indices

nx = 0:length(x)-1;
ny = 0:length(y)-1;
l = -(length(y)-1):(length(x)-1);

% Plot signals

figure;

% Signal x[n]
subplot(3,1,1);
stem(nx, x, 'filled', 'LineWidth', 1.5);
grid on;
title('Signal x[n]');
xlabel('n');
ylabel('Amplitude');

% Signal y[n]
subplot(3,1,2);
stem(ny, y, 'filled', 'LineWidth', 1.5);
grid on;
title('Signal y[n]');
xlabel('n');
ylabel('Amplitude');

% Cross-Correlation
subplot(3,1,3);
stem(l, r, 'filled', 'LineWidth', 1.5);
grid on;
title('Cross-Correlation  $r_{xy}[l]$ ');
xlabel('Lag (l)');
ylabel('Amplitude');

```

%Unit Sequence

```
clc;
```

```
clear;
```

```
close;
```

```
n=-5:5;
```

```
%Impulse Signal
```

```
x1=(n==0);
```

```
%Unit step signal
```

```
x2=(n>=0);
```

```
%Ramp signal
```

```
x3=n.*(n>=0);
```

```
%Impulse
```

```
subplot(3,1,1);
```

```
stem(n,x1,"filled",'LineWidth',1.5);
```

```
%Unit
```

```
subplot(3,1,2);
```

```
stem(n,x2,"filled",'LineWidth',1.5);
```

```
%Ramp
```

```
subplot(3,1,3);
```

```
stem(n,x3,"filled",'LineWidth',1.5);
```